

Multiple Choice

1. **Answer:** A

Options: A) 0.7%; B) 3%; C) 0.1%; D) 1.5%; E) 1%

Note: The correct answer of 0.7% for the percentage of angular magnification when viewing an object at 33 cm through a 5 mm thick pane of glass is derived from the minimal impact of the glass's thickness on the angular size of the object, which results in a very small change in the perceived angle, thus leading to a low percentage magnification.

2. **Answer:** A

Options: A) 3.0×10^{-32} ; B) 5.5×10^{-33} ; C) 1.2×10^{-30} ; D) 8.9×10^{-36} ; E) 2.3×10^{-30}

Note: The correct answer is derived from the de Broglie wavelength formula, $\lambda = h/(mv)$, where h is Planck's constant ($6.626 \times 10^{-34} Js$), m is the mass of the tennis ball in kg ($0.057 kg$), and v is the speed of the ball in m/s. This illustrates the wave-particle duality of matter at macroscopic scales.

2. **Answer:** A

Options: A) 1200 lb; B) 1800 lb; C) 2000 lb; D) 1000 lb; E) 2400 lb

Note: The necessary centripetal force is calculated using the formula $F_c = \frac{mv^2}{r}$, where m is the mass of the car in slugs (converted from pounds), v is the speed in feet per second, and r is the radius of the curve, resulting in a centripetal force of 1200 lb.

3. **Answer:** E

Options: A) more fragile; B) shorter; C) heavier; D) equal in height; E) taller

Note: A barometer that uses water instead of mercury will be taller because water has a lower density than mercury, requiring a greater height of water column to balance atmospheric pressure.

4. **Answer:** A

Options: A) asteroids would orbit with a period that is three times that of Jupiter; B) asteroids would orbit with a period three times that of Mars; C) asteroids would orbit with a period twice that of Mars; D) asteroids would orbit with a period twice that of Jupiter; E) A and B

Note: Kirkwood gaps occur in the asteroid belt at positions where the orbital periods of asteroids resonate with the gravitational influence of Jupiter, specifically where asteroids have orbital periods that are integer fractions of Jupiter's, such as one-third, leading to a destabilizing gravitational interaction and resulting in fewer asteroids in those regions.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The displacement of a hiker is defined as the shortest straight-line distance between the starting and ending points, and it does not equate to half the total distance covered, which includes all segments of the hiker's path.

7. **Answer:** False

Options: A) True; B) False

Note: One astronomical unit (AU) is actually defined as approximately 149.6 million kilometers, not 170 million kilometers.

8. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the relationship between actual brightness (intrinsic luminosity) and apparent brightness (observed brightness) follows the inverse square law, allowing for the calculation of distance based on the ratio of these two measurements.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because shorter wavelengths, such as blue light, refract more than longer wavelengths, like green light, when passing through a prism due to the dependence of refraction on wavelength as described by Snell's law.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because, under constant acceleration due to gravity, the distance covered in each second of free fall increases linearly, resulting in a total distance of 2 h during the second second, where h is the distance fallen in the first second.

Long Form Response

11. **Answer:** When a proton falls under the influence of Earth's gravity, it accelerates and gains kinetic energy as it descends. If we assume the proton converts all of this kinetic energy into a quantum of light, we can use the relationship between energy and frequency given by Planck's equation, $E = hf$, where E is energy, h is Planck's constant, and f is frequency. The kinetic energy gained by the proton can be expressed as $E = mgh$, where m is the mass of the proton, g is the acceleration due to gravity, and h is the height fallen. By substituting the values (mass of the proton $\approx 1.67 \times 10^{-27} \text{ kg}$, $g \approx 9.81 \text{ m/s}^2$), and calculating the energy, we find that the frequency of the emitted photon is approximately $2.2 \times 10^{14} \text{ sec}^{-1}$. *This result illustrates the conversion of gravitational potential energy*

Note: correct because it accurately applies the principles of conservation of energy, relating the gravitational

11. **Answer:** To calculate the frequency of the hydrogen transition from $n = 101$ to $n = 100$, we can use the Rydberg formula, which relates the wavelengths of emitted or absorbed light to the energy levels of an atom. The energy difference between these two levels can be expressed as $E = E_n - E_m = -13.6 \text{ eV} (1/n^2 - 1/m^2)$. For this transition, substituting $n = 101$ and $m = 100$ yields an energy difference of $E = -13.6 \text{ eV} (1/101^2 - 1/100^2)$, resulting in a frequency of approximately $6.5 \times 10^9 \text{ Hz}$ when converted using the relation $E = hf$, where h is Planck's constant. This transition, while involving high quantum numbers, is significant as it demonstrates the quantized nature of atomic energy levels and provides insights into atomic structure and the behavior of electrons in hydrogen.

Note: The answer correctly applies the Rydberg formula to calculate the energy difference between the hydrogen atom's energy levels $n = 101$ and $n = 100$, ultimately deriving the corresponding frequency of the emitted or absorbed photon, which is significant in understanding atomic transitions and the quantization of energy levels.

End of Answer Key